

Spatial Distribution of Bidirectional Charging Stations by Simulation and Evaluation of Traffic and Grid Data

Achim BAUMGÄRTNER¹, Erik WEIGAND¹,
Gian Bruno TRUÖL^{1,2}, Mirko SONNTAG^{1,2}

¹University of Applied Sciences Esslingen, Esslingen University, Germany

²Fraunhofer Application Center Keim, Esslingen, Germany

acbait01@hs-esslingen.de, erwewi01@hs-esslingen.de,
gian-bruno.truoel@hs-esslingen.de, mirko.sonntag@hs-esslingen.de

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Abstract. This paper investigates how bidirectional charging stations can be distributed across an urban area by combining traffic simulation with power-grid indicators. We model charging demand as a spatiotemporal process and evaluate candidate locations with respect to accessibility, waiting time, and local grid stress. The proposed workflow integrates mobility traces, station utilization forecasts, and feeder-level constraints to identify robust deployment strategies. Results indicate that jointly optimizing traffic and grid criteria can reduce peak congestion while improving charging availability in high-demand districts.

Keywords: Bidirectional charging, traffic simulation, real-time modeling, neural networks, Python

1. Introduction

The transition to electric mobility is shifting infrastructure planning from static placement rules to data-driven decision support. Bidirectional charging adds further complexity because stations are not only energy consumers but also potential distributed flexibility assets. As a result, charging station siting must be evaluated with both transportation efficiency and grid stability in mind.

1.1 Motivation

Urban charging demand is highly uneven across time and location, which can cause queues at popular stations while other sites remain under-used. At the same time, local transformer and feeder limits may prevent straightforward expansion in high-demand zones. A coordinated planning approach is therefore required to balance user convenience and grid feasibility.

1.2 Problem Statement

Given a set of candidate charging locations, we aim to determine a deployment configuration that minimizes travel and waiting times while respecting grid capacity constraints. The challenge is multi-objective and dynamic, since mobility demand and feeder loading change throughout the day. We formulate this as a constrained optimization problem informed by simulation outputs from both traffic and grid models.

1.3 Contributions

This paper contributes a practical and reproducible workflow for planning bidirectional charging infrastructure with coupled traffic and grid evaluation. The approach is designed to support municipal planning and utility stakeholders with transparent trade-off analysis.

The main contributions of this work are as follows:

- A simulation-based siting pipeline that jointly evaluates mobility accessibility and distribution-grid impact.
- A multi-criteria scoring framework for comparing candidate station portfolios under realistic demand variability.
- A reference implementation in Python to support reproducible experiments and scenario-based sensitivity studies.

1.4 Paper Structure

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 describes the proposed methodology. Section 4 presents experimental results. Section 5 discusses findings and limitations. Section 6 concludes the paper.

2. Related Work

Prior work on charging infrastructure planning typically focuses on either mobility demand coverage or electrical grid constraints in isolation. More recent studies combine transportation simulation and power-system analysis, but integrated frameworks for bidirectional charging remain limited.

2.1 Traffic-Aware Siting

Traffic-aware siting methods estimate charging demand from origin-destination flows, activity patterns, and route choices. These approaches improve station accessibility and reduce detours, yet they may overlook local network bottlenecks in the distribution grid.

2.2 Grid-Aware Siting

Grid-aware planning approaches prioritize feeder hosting capacity, voltage stability, and peak-load mitigation. While they improve technical feasibility, they can produce suboptimal user experience if travel behavior is not modeled with sufficient detail.

3. Methodology

Our methodology links a mesoscopic traffic simulation with a distribution-grid assessment model and a ranking module for candidate station layouts. Each scenario is evaluated over multiple demand profiles to capture variability and uncertainty.

3.1 System Overview

The workflow consists of four stages: data preprocessing, demand simulation, grid impact estimation, and multi-criteria scoring. Inputs include road-network topology, travel demand traces, candidate station coordinates, and feeder-level electrical limits. Outputs are scenario-level KPIs that quantify accessibility, utilization, and network stress.

3.2 Model Architecture

Demand prediction is represented by a spatiotemporal model that maps traffic states and contextual features to charging events. A separate grid module estimates nodal loading and transformer utilization from aggregated charging schedules. Both modules are coupled through iterative updates so that mobility and grid effects remain consistent.

The loss function is defined as:

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i(\theta))^2 + \lambda \|\theta\|_2^2 \quad (1)$$

where y_i denotes the ground-truth label and $\hat{y}_i$ the model prediction.

3.3 Training Procedure

Model parameters are trained with mini-batch optimization and early stopping based on validation loss. We perform repeated runs with different random seeds to reduce variance in performance estimates. Hyperparameters are selected using a bounded search space to balance accuracy and computational cost.

Table 1: Model configuration hyperparameters

Parameter	Value	Search Space
Learning Rate	0.001	$[10^{-4}, 10^{-2}]$
Batch Size	64	$\{32, 64, 128\}$
Hidden Layers	3	$\{2, 3, 4\}$
Dropout Rate	0.2	$[0.1, 0.5]$
Epochs	100	–

4. Experiments

We evaluate the proposed planning approach on representative urban scenarios with varying fleet sizes and charging behaviors. The experiments compare baseline siting heuristics against our integrated optimization strategy.

4.1 Experimental Setup

All experiments are executed on synchronized traffic and grid snapshots with 15-minute resolution. We run each scenario over multiple simulated days to account for temporal fluctuations in demand and network loading.

4.2 Datasets

The traffic dataset includes synthetic trip chains calibrated to observed commuting and leisure patterns. Grid data contains feeder topology, transformer ratings, and operational constraints provided at district level.

4.3 Evaluation Metrics

We report accessibility metrics (average detour time and queueing delay), utilization metrics (station occupancy and throughput), and grid metrics (peak loading and overload frequency). In addition, we compute a composite score to summarize planning trade-offs across objectives.

4.4 Results

The integrated method consistently improves accessibility while reducing critical grid stress events relative to single-domain baselines. Benefits are strongest during evening peaks, where

coordinated siting shifts demand away from constrained feeders. Overall, results suggest that multi-domain optimization yields more robust infrastructure portfolios.



Figure 1: Comparison of model accuracy on the test dataset.

5. Discussion

The results highlight the importance of coupling transportation demand with local grid constraints during station planning. In particular, locations that appear optimal from a pure mobility perspective may create unacceptable stress at nearby substations.

5.1 Interpretation of Results

Our findings indicate that balanced station portfolios can improve user service levels without relying on costly grid reinforcement in every high-demand area. The strongest gains arise when moderate detour increases are traded for substantial reductions in feeder overload risk.

5.2 Limitations

This study relies on scenario assumptions for travel behavior, charging preferences, and controllability of vehicle-to-grid operation. Real-world deployment may introduce additional uncertainties, such as policy constraints and user acceptance effects, that are not fully captured in the current model.

6. Conclusion

This paper presented a simulation-driven method for distributing bidirectional charging stations under joint traffic and grid criteria. The

evaluation demonstrates that coordinated planning can improve both charging accessibility and electrical reliability.

6.1 Future Work

Future work will include stochastic optimization under forecast uncertainty and integration of dynamic pricing signals. We also plan to validate the framework with larger real-world datasets and pilot deployments in mixed urban-suburban regions.

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